Create a class Vehicle with attributes like make and model. Implement subclasses Car and ElectricCar. Car should have an attribute number of doors, while ElectricCar should have an attribute battery capacity. Car & ElectricCar must be maintained in separate packages. Implement interface with a display method. Use static methods to count the number of vehicles created.

Create an interface LibraryItem with methods like borrowItem() and returnItem(). Implement classes Book and DVD that implement LibraryItem. Book should have attributes like title and author while DVD should have attributes

like title, director, and duration. Book & DVD must be maintained in separate packages. Implement a Library class that manages a collection of LibraryItem objects

Create an abstract class Person with attributes like name and age. Implement subclasses Student and Professor. Student should have attributes like student ID and course, while Professor should have attributes like employee ID and department. Implement interface with a display method. The Student & Professor must be maintained in separate packages. Implement methods to create 2 batches of Students and a Professor for each batch. Display details for each batch.